Week 7 - Graded Assignment 7

The due date for submitting this assignment has passed.
Due on 2026-04-01, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-01, 07:55 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1 point
Consider the vector spaces V and functions $\langle .,. \rangle : V \times V \to R$ defined as follows:

i) $V = R^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_1 - x_1 y_2 + 2x_2 y_2$.
ii) $V = M_{2\times 2}(R)$ and $\langle A, B \rangle = Tr(AB)$,
where $Tr(M)$ denotes the trace of a matrix M, i.e., the sum of the diagonal elements of M.
iii) $V = M_{2\times 1}(R)$ and $\langle A, B \rangle = Tr(AB^t)$,
where $Tr(X)$ denotes the trace of a matrix X, i.e., the sum of the diagonal elements of X. Y^t denotes the transpose of matrix Y.
iv) $V = R^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 + x_2 y_1$.

- [x] (i) is an inner product.
- [x] (ii) is an inner product.
- [ ] (iii) is an inner product.
- [ ] (iv) is an inner product.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(i) is an inner product.
(iii) is an inner product.

1 point
Consider two linear transformations T and S from $R^2 \to R^2$ defined as $T(x, y) = (2x + y, x + y)$ and $S(x, y) = (x + cy, x + 2y)$. Let A and B be matrix representations of linear transformations T and S with respect to the standard bases of R^2 respectively.
Consider the following statements:

- **P:** If $c = 1$, then A and B are similar matrices.
- **Q:** If $c = 2$, then A and B are similar matrices.
- **R:** If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$.
- **S:** If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.
- **T:** If $c = 1$, then there are infinitely many P satisfying the equation $P^{-1}AP = B$.

Which of the following options are true?

- [x] P is true but Q is false.
- [ ]

Both P and Q are true.

- [ ] Both R and S are true.
- [x] R is false but S is true.
- [x] T is true.

Yes, the answer is correct.
Score: 1
Accepted Answers:
P is true but Q is false.
R is false but S is true.
T is true.

1 point

Let $\langle ., . \rangle$ denote the standard inner product on R^2, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$. Which one of the following options is (are) true for the vector $\gamma \in R^2$, such that $\langle \alpha, \gamma \rangle = 4$ and $\langle \beta, \gamma \rangle = 8$, where $\alpha = (3, 1)$ and $\beta = (6, 2)$.

- [ ] No such γ exists.
- [x] There are infinitely many such vectors which satisfy the properties of γ.
- [ ] γ is unique in R^2.
- [x] Any vector in the set $\{(t, 4 - 3t) \mid t \in R\}$ satisfies the properties of γ.
- [ ] $(1, 1)$ is the only vector which satisfies the properties of γ.

Yes, the answer is correct.
Score: 1
Accepted Answers:
There are infinitely many such vectors which satisfy the properties of γ.
Any vector in the set $\{(t, 4 - 3t) \mid t \in R\}$ satisfies the properties of γ.

1 point

Let $\langle ., . \rangle$ denote the standard inner product on R^2 , i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ and let $v \in R^2$. Consider a linear transformation $T_v : R^2 \rightarrow R$ defined as:

$$T_v(u) = \langle u, v \rangle,$$

where $v \in R^2$. Which of the following options is (are) true for T_v?

- [ ] T_v is one-one for all $v \neq 0 \in R^2$.
- [x] T_v is onto for all $v \neq 0 \in R^2$.
- [ ] T_v is onto for all $v \in R^2$.
- [x] T_v is not one-one for every $v \in R^2$.
- [x]

There exists a $v \in R^2$ such that T_v is not onto.

Yes, the answer is correct.
Score: 1
Accepted Answers:
T_v is onto for all $v \neq 0 \in R^2$.
T_v is not one-one for every $v \in R^2$.
There exists a $v \in R^2$ such that T_v is not onto.

1 point
Let A and B be square matrices of the same order n. Which of the following statements are sufficient to conclude that A is equivalent to B.

- [ ] det(A) = det(B).
- [x] Nullity(A) = Nullity(B).
- [ ] The set of columns of A, and the set of columns of B are both linearly dependent subsets of R^n.
- [x] There exists a matrix C such that A is equivalent to C, and B is equivalent to C.
- [x] There exist invertible matrices P and Q of order n such that $PA = BQ$.

Yes, the answer is correct.
Score: 1
Accepted Answers:
Nullity(A) = Nullity(B).
There exists a matrix C such that A is equivalent to C, and B is equivalent to C.
There exist invertible matrices P and Q of order n such that $PA = BQ$.

1 point
Let L and L' be affine subspaces of R^3, where $L = (0,1,1) + U$ and $L' = (0,1,0) + U'$, for some vector subspaces U and U' of R^3. Let a basis for U be given by $\{(1,1,0),(1,0,1)\}$ and a basis for U' be given by $\{(1,0,0)\}$. Suppose there is a linear transformation $T: U \to U'$ such that $(1,0,1) \in \ker(T)$ and $T(1,1,0) = (1,0,0)$. An affine mapping $f: L \to L'$ is obtained by defining $f((0,1,1)+u) = (0,1,0) + T(u)$, for all $u \in U$. Which of the following options are true?

- [x] $L = \{(x, y+1, x-y+1) \mid x, y \in R\}$.
- [ ] $L' = \{(x, y+1, 0) \mid x, y \in R\}$.
- [ ] $L = \{(x-y, y+1, x-y+1) \mid x, y \in R\}$.
- [ ] $L = \{(x, x+1, y+1) \mid x, y \in R\}$.
- [x] $f(x, y+1, x-y+1) = (y, 1, 0)$
- [ ] $f(x-y, y+1, x-y+1) = (x, y+1, 0)$
- [ ] $f(x, x+1, y+1) = (y, 1, 0)$
- [ ] $f(x, y+1, x-y+1) = (0, 1, y)$

Yes, the answer is correct.
Score: 1

Accepted Answers:

$L = \{(x, y+1, x-y+1) \mid x, y \in R\}$.

$f(x, y+1, x-y+1) = (y, 1, 0)$

Let θ be the angle between the vectors $u = (4, 7, 3)$ and $v = (1, 2, -6)$, then what will be the value of $cos(\theta)$?

0

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 0

1 point

Consider a basis

$$\{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\}$$

of R^3 with usual inner product. Suppose $v = (x, y, \frac{3x+y}{5}) \in V$ is written as $v = c_1 v_1 + c_2 v_2 + c_3 v_3$, such that $c_1 + c_2 = 4$. What will be the value of c_3?

2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2

1 point

Let $V = R^2$ be a vector space. Consider two inner products $\langle .,. \rangle_1$ and $\langle .,. \rangle_2$ on V defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1 x_2 - x_1 y_2 - x_2 y_1 + 4 y_1 y_2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3 x_1 x_2 + 2 y_1 y_2.$$

If $\langle (a, b), (4, 3) \rangle_1 = 35$ and $\langle (a, b), (4, 3) \rangle_2 = 60$, then find the value $a + 2b$.

11

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 11

1 point

Let $V = R^3$ be the inner product space with usual inner product. If θ is the angle between $(15, 8, 18)$ and (a, b, c) where $15a + 8b + 18c = 0$ and $(a^2 + b^2 + c^2) \neq 0$, then find the value of $\cos\theta$.

0

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 0

1 point

Consider a vector $v = (x - 20, 2, 1) \in R^3$ v = (x − 20, 2, 1) ∈ R3. Find the value of xx so that the length of the vector vv is minimum.

20

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 20
1 point

Consider the following subsets of R^3 R3, with $\langle , \rangle$⟨ , ⟩ used to denote the standard dot product on R^3 R3, and $\| \|$‖ ‖ used to denote the standard length function.
$S_1 = \{v \in R^3 \mid \langle v, (0,0,1) \rangle = 2, \|v\| = 1\}$ $S_2 = \{u \in R^3 \mid u \in \text{Span}(\{(1,1,1)\}), \|u\| = 1\}$S1 = {v ∈ R3 | ⟨v, (0, 0, 1)⟩ = 2, ‖v‖ = 1}S2 = {u ∈ R3 | u ∈ Span({(1, 1, 1)}), ‖u‖ = 1}
If n_1 n1 and n_2 n2 denote the number of elements in S_1 S1 and S_2 S2, respectively, then find the value of $n_1 - n_2$ n1 − n2.

-2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) -2
1 point